

PAPER The Kronos Strategic-Materials Platform: An Integrated Spherical-Tokamak Breeder and D–³He Tandem-Mirror Burner Architecture with a Helium-3 Flywheel

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Abstract

Compact fusion is usually presented as a single machine chasing net electricity. Kronos is instead a *platform* of two machines with one physics record: a compact negative-triangularity spherical-tokamak *breeder* (Hyperion) that makes strategic materials, and a deuterium–helium-3 tandem-mirror *burner* (deployed as Aegis and MetroVolt) that makes power. We give a single, self-consistent formal treatment of the platform. The breeder operates at gain $Q = 3.076$ with fusion power $P_{\text{fus}} = 85.04$ MW and plasma current $I_p = 9.66$ MA behind a 16.84 T centrepost, supplying tritium along a net breeding ladder that clears self-sufficiency ($\text{TBR}_{\text{net}} = 1.05$) and reaches an advanced ceiling of 1.42; the burner reaches engineering gain $Q_E = 1.318$ at a neutron energy fraction of only 5.44%, recovering charged-particle power through a direct converter at $\eta_{\text{DEC}} = 0.49$ behind a distinct 26.49 T plug coil. We derive the two power balances, the breeding neutronics, and—as the paper’s central contribution—the *helium-3 flywheel*: a formal inventory model in which surplus tritium is banked and decays to ³He with a 12.32 yr half-life, so the strategic co-product is released at a rate that approaches the tritium surplus only over an e -folding time of $\tau \approx 17.8$ yr. This slow discharge is decisive: matching a burner’s 170.2 kg yr^{-1} helium-3 demand from banked-tritium decay would require 45–192 breeders per burner, so burner-scale helium-3 does not close indigenously and is lunar-gated around 2038–40, while each breeder’s $\sim 0.15\text{--}0.9 \text{ kg-T yr}^{-1}$ of ³He remains a strategic co-product rather than bulk fuel. Every headline quantity is a frozen anchor in the Kronos Physics De-Risking Register, and the whole design set was produced and independently re-run on an NVIDIA GPU high-performance-computing campaign. We state closure in two explicit tiers throughout: all three products close on the frozen physics, and the distance from closed-on-model to closed-demonstrated is a small number of named experiments, not open physics.

Keywords: fusion platform; spherical-tokamak breeder; tandem-mirror burner; helium-3; tritium breeding; direct energy conversion; two-tier closure

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1 Introduction

1.1 From a single machine to a platform

The dominant framing of magnetic fusion is a race to net electricity from one machine. That framing has organised the field productively for decades, from the Lawson power-balance criterion [1] through the ITER physics basis [2], but it obscures a fact that the neutron economy of fusion makes plain: a deuterium–tritium plasma is, before it is a power source, a *strategic-materials factory*. A single 14 MeV neutron breeds tritium, transmutes lithium, multiplies in beryllium, and activates target nuclei into medical and industrial isotopes. Kronos is built on that observation. It is not one machine but a platform of two (figure 1): a compact spherical-tokamak *breeder* that makes strategic materials, and a deuterium–helium-3 tandem-mirror *burner* that makes power, sharing a REBCO high-field magnet basis, an autonomous control architecture, and one frozen physics de-risking record [25].

This paper is the integrated overview of that platform. Each subsystem has a dedicated companion paper in the Kronos 2026 series; here we assemble the whole architecture into a single formal treatment, derive its governing balances, and—most importantly—give the mathematics of the object that links the two machines without closing between them, the *helium-3 flywheel*. Every headline number in this paper is a frozen anchor in the de-risking register [25]; the serial identifiers (e.g. BR-OUT-4) are quoted inline so each quantity is auditable.

1.2 Prior art

The breeder side descends directly from the spherical-tokamak fusion-nuclear-science-facility and pilot-plant studies of Menard *et al.* [5] and from the compact high-field, demountable-magnet line opened by ARC [6] and carried into REBCO practice by SPARC [7] and by the 45.5 T all-superconducting record of Hahn *et al.* [8]. Its confinement strategy is negative triangularity, whose reactor-relevant, edge-localised-mode-free performance was established on DIII-D by Austin *et al.* [9] and reviewed by Marinoni *et al.* [10]; its density envelope is bounded by the Greenwald limit [4]; its breeding-blanket physics follows the tritium fuel-cycle and self-sufficiency analyses of Abdou *et al.* [11]. The burner side descends from the tandem-mirror line: the ambipolar-potential confinement idea of Dimov *et al.* and Fowler and Logan [16, 15], the collisional-loss theory of Pastukhov [14], the mirror-program synthesis of Post [17], and its modern high-field revival in the Wisconsin HTS Axisymmetric Mirror (WHAM) [18]. Its fuel choice rests on the advanced-fuel case of Kulcinski and Santarius [19] and the confinement-requirement reviews of Nevins [20], disciplined by Rider’s limits on non-equilibrium recovery [21]; its power train is the direct converter of Moir and Barr [22]; and its bulk fuel is the lunar helium-3 resource identified by Wittenberg, Santarius and Kulcinski [23] and mapped by later remote sensing [24].

1.3 Contribution

What is missing from that literature is a single architecture that treats a breeder and a burner as *one* program with quantitatively coupled fuel cycles, and that states honestly where the strategic loop closes and where it does not. This paper contributes exactly that. We (i) give a unified 0-D power balance for the two very different machines and their shared magnet basis (§2); (ii) derive the net tritium-breeding ladder and the per-unit surplus law that feeds the flywheel (§3); (iii) formalise the *helium-3 flywheel* as a banked-tritium decay inventory with an explicit time constant, and prove from it that indigenous helium-3 cannot fuel a burner fleet (§4); (iv) close the burner power balance through the Boltzmann/Pastukhov plug potential and the direct converter (§5); (v) describe the NVIDIA GPU high-performance-computing campaign that produced and independently re-ran the frozen result set (§6); and (vi) present the results, a quantitative comparison to published benchmarks, one honest open gate, and a two-tier closure statement (§§7–10).

The Kronos platform: one breeder of strategic materials, one burner of power

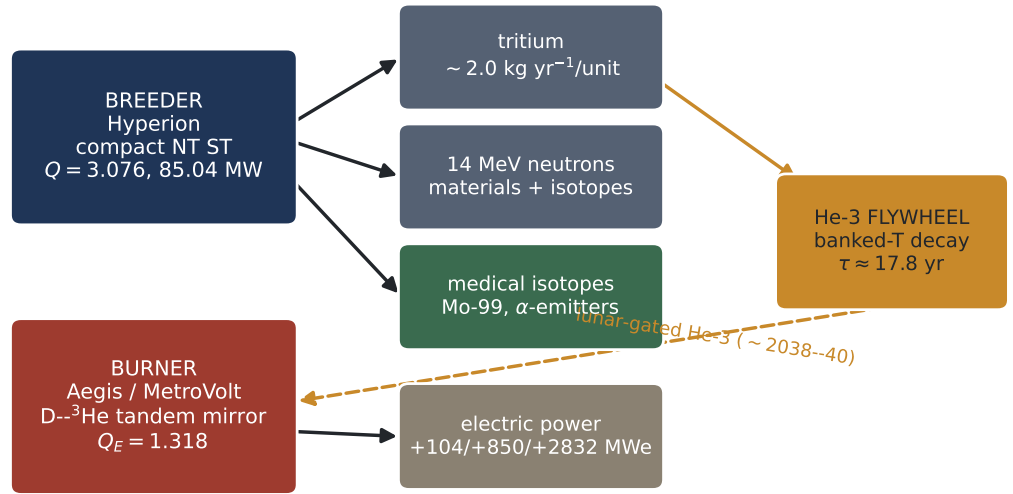


Figure 1. (Schematic.) The Kronos platform as one architecture. The breeder (Hyperion) supplies four strategic-material products; the burner (Aegis/MetroVolt) supplies power. Surplus tritium charges the helium-3 flywheel, which discharges slowly by radioactive decay; burner-scale helium-3 is lunar-gated. No data values are encoded; the figure shows structure only.

2 Governing equations and the formal problem

We treat the platform as a set of coupled 0-D balances that share nuclear and magnet constants. This section states the two machine balances; §§3–4 develop the tritium/helium-3 fuel cycle that couples them.

2.1 Fusion reactivities and the two fuel choices

Both machines are driven by Maxwellian fusion reactivities $\langle\sigma v\rangle(T_i)$, for which we use the Bosch–Hale parametrisation [3]

$$\langle\sigma v\rangle = C_1 \theta \sqrt{\frac{\xi}{m_r c^2 T_i^3}} e^{-3\xi}, \quad \theta = \frac{T_i}{1 - \frac{T_i(C_2 + T_i(C_4 + T_i C_6))}{1 + T_i(C_3 + T_i(C_5 + T_i C_7))}}, \quad \xi = \left(\frac{B_G^2}{4\theta}\right)^{1/3}, \quad (1)$$

with reaction-specific Gamow constant B_G , reduced mass $m_r c^2$ and fit coefficients $C_1 \dots C_7$. Evaluating equation (1) for the D–T, D– ^3He and D–D channels (figure 2) exposes the two design choices at a glance. The breeder runs D–T near the reactivity knee at $T_i \sim 15\text{--}20\text{ keV}$, where $\langle\sigma v\rangle_{\text{DT}}$ is large and the source is deliberately neutron-rich; the burner runs D– ^3He at $T_i \sim 85\text{ keV}$, where the reaction is charged-particle dominated and the parasitic neutron channels are suppressed. The two machines are thus not variants of one design point but occupy opposite ends of the reactivity plane.

2.2 Breeder power balance and gain

The breeder is a compact negative-triangularity spherical tokamak with major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$, triangularity $\delta = -0.30$, on-axis field $B_0 = 8$ T and plasma current $I_p = 9.66$ MA. With $n_D = n_T = n_i/2$ the volumetric D–T fusion power density is

$$p_{\text{fus}} = \frac{1}{4} n_i^2 \langle \sigma v \rangle_{\text{DT}}(T_i) E_{\text{DT}}, \quad E_{\text{DT}} = 17.59 \text{ MeV}, \quad (2)$$

of which the fraction $f_n^{\text{DT}} = E_n/E_{\text{DT}} = 14.06/17.59 \simeq 0.80$ is carried by the neutron. Integrating over the plasma volume V gives $P_{\text{fus}} = \int_V p_{\text{fus}} dV = 85.04$ MW at the frozen operating point (BR-OUT/HX anchors). Steady state is a global energy balance between external heating P_{ext} , alpha heating $P_\alpha = \frac{1}{5} P_{\text{fus}}$, and transport plus radiation losses,

$$P_{\text{ext}} + P_\alpha = \frac{W}{\tau_E} + P_{\text{rad}}, \quad W = \frac{3}{2} \int_V (n_i T_i + n_e T_e) dV, \quad (3)$$

with W the thermal stored energy and τ_E the energy confinement time. The plasma gain follows as

$$Q \equiv \frac{P_{\text{fus}}}{P_{\text{ext}}} = 3.076, \quad (4)$$

a deliberately *subcritical*, driven operating point ($Q < Q_{\text{ign}}$): the breeder is a materials factory whose figure of merit is source strength and breeding, not ignition. The negative-triangularity edge is what makes this Q accessible without an H-mode pedestal, by suppressing ion- and electron-scale turbulence into a quiet subcritical core [9, 10]; the confinement companion establishes the turbulence quiescence at the operating point (doi:10.5281/zenodo.22645722). The corresponding 14 MeV source strength is

$$S_n = \frac{f_n^{\text{DT}} P_{\text{fus}}}{E_n} \approx \frac{0.80 \times 85.04 \text{ MW}}{14.06 \text{ MeV}} \approx 3.0 \times 10^{19} \text{ n s}^{-1}, \quad (5)$$

the quantity that drives every downstream neutronic product.

2.3 The shared magnet basis: two distinct magnets

Both machines are wound from no-insulation REBCO coated conductor, but they use *two physically distinct magnets* that must never be conflated. The breeder centrepost governs at a peak conductor field of 16.84 T; the burner plug coil is a separate, higher-field magnet at 26.49 T (figure 8). On each coil's load line the current-sharing temperature margin $\Delta T_{\text{cs}} = T_{\text{cs}} - T_{\text{op}}$ is the protective observable; at design current the centrepost holds $\Delta T_{\text{cs}} \approx 49.5$ K at load fraction $I_{\text{op}}/I_c \approx 0.27$, and the plug $\Delta T_{\text{cs}} \approx 37.1$ K at ≈ 0.43 . Keeping the two fields separate is a load-bearing element of the magnet-integrity case; the plug-coil companion carries the three-dimensional structural analysis of the 26.49 T coil (doi:10.5281/zenodo.22645915).

3 Tritium breeding and the per-unit surplus law

The first strategic product is tritium. The net tritium-breeding ratio is the volume integral of the tritium-producing reaction rate per source neutron,

$$\text{TBR}_{\text{net}} = \frac{1}{S_n} \int_V \int_0^\infty n_{\text{Li}}(\mathbf{r}) [\sigma_6(E) + \sigma_7(E)] \phi(\mathbf{r}, E) dE dV \cdot M f_{\text{cov}} - L, \quad (6)$$

where σ_6 is the ${}^6\text{Li}(n, \alpha)\text{T}$ cross-section, σ_7 the ${}^7\text{Li}(n, n'\alpha)\text{T}$ threshold channel, ϕ the neutron flux, M the beryllium/ Be_{12}Ti multiplication factor, f_{cov} the blanket coverage fraction, and L the leakage/parasitic-absorption loss. The compact spherical-tokamak geometry caps inboard coverage through the centrepost, so f_{cov} is the dominant lever. Monte-Carlo transport with OpenMC [12] over the frozen blanket recipe yields the breeding ladder of figure 3: the as-drawn recipe gives $\text{TBR}_{\text{net}} = 0.742$ (a net consumer), the solid-beryllium coverage lever raises it to 1.123/1.189/1.215 at $f_{\text{cov}} = 0.85/0.90/0.92$ (BV-05), a CAD-faithful three-dimensional model with real ports returns 1.120 (single-null, BV-05B), and the advanced blanket reaches a ceiling of 1.42 (BR-L2-A7u). The self-sufficiency line $\text{TBR}_{\text{net}} = 1.05$ is cleared with margin; the cross-library net-TBR case and the breeding-fleet argument are carried by companions (doi:10.5281/zenodo.22645706, doi:10.5281/zenodo.22645746).

The quantity that feeds the flywheel is the *surplus* tritium rate. With tritium burn rate $\dot{N}_{\text{burn}} = 13.1 \text{ g d}^{-1} = 4.78 \text{ kg yr}^{-1}$ (BR-SX-17), conservation of the breeding cycle gives the per-unit surplus law

$$\dot{N}_{\text{surplus}} = (\text{TBR}_{\text{net}} - 1) \dot{N}_{\text{burn}} \quad (7)$$

which reproduces the frozen anchors exactly: $\dot{N}_{\text{surplus}} = 0.89/1.62/3.82 \text{ kg-T yr}^{-1}$ at $\text{TBR}_{\text{net}} = 1.186/1.34/1.80$ (BR-OUT-1, BR-OUT-4). Equation (7) is the input to the flywheel model of the next section.

4 The helium-3 flywheel

Helium-3 is the strategic product that links the two machines—and the object whose dynamics decide that the loop does not close indigenously. We formalise it as an inventory model.

4.1 Banked-tritium decay dynamics

Surplus tritium that is not sold or fed forward is *banked*, and tritium decays to helium-3, $\text{T} \rightarrow {}^3\text{He} + e^- + \bar{\nu}_e$, with half-life $t_{1/2} = 12.32 \text{ yr}$ and decay constant $\lambda = \ln 2/t_{1/2} = 0.0563 \text{ yr}^{-1}$. Writing $N_T(t)$ for the banked tritium inventory (per unit) charged at the constant surplus rate $\sigma \equiv \dot{N}_{\text{surplus}}$ of equation (7), the reservoir obeys the linear balance

$$\frac{dN_T}{dt} = \sigma - \lambda N_T, \quad (8)$$

whose solution with $N_T(0) = 0$ is

$$N_T(t) = \frac{\sigma}{\lambda} (1 - e^{-\lambda t}), \quad N_T^\infty = \frac{\sigma}{\lambda} = \sigma \tau, \quad \tau \equiv \frac{1}{\lambda} = 17.8 \text{ yr}. \quad (9)$$

The helium-3 *production* rate is the decay rate of the banked reservoir,

$$\dot{N}_{3\text{He}}(t) = \lambda N_T(t) = \sigma (1 - e^{-\lambda t}) \xrightarrow[t \rightarrow \infty]{} \sigma, \quad (10)$$

so in equilibrium the helium-3 production rate equals the tritium surplus rate—one helium-3 nucleus per decayed tritium—but it *approaches* that equilibrium only over the *e*-folding time $\tau \approx 17.8 \text{ yr}$, with a half-response at exactly $t_{1/2} = 12.32 \text{ yr}$ (figure 4). This is the flywheel: strategic material stored as tritium and released as helium-3 with a large mechanical-analogue time constant. The cumulative helium-3 delivered by time t is $\int_0^t \lambda N_T dt' = \sigma[t - \tau(1 - e^{-\lambda t})]$. Independently of the banked reservoir, each unit harvests $\sim 154 \text{ g yr}^{-1}$ of helium-3 in situ (113 g from in-vessel tritium decay plus 41 g from the D–D branch), a steady plasma inventory of $\sim 5 \times 10^{17}$ atoms held at 0.012% of n_e without dilution (BR-CX-16, BR-CX-29). At equilibrium the banked path delivers up to $\sim 0.9 \text{ kg-T yr}^{-1}$ per unit (BR-OUT-2).

4.2 The doubling limit

The same decay term bounds how fast a tritium reservoir can grow to start new units. If the surplus is reinvested, the reservoir grows as $\dot{N}_T = (\text{TBR}_{\text{net}} - 1)\dot{N}_{\text{burn}} - \lambda N_T - \ell N_T$, with ℓ the process-loss rate; the inventory doubling time is

$$t_{\text{double}} = \frac{\ln 2 N_{T,0}}{(\text{TBR}_{\text{net}} - 1)\dot{N}_{\text{burn}} - (\lambda + \ell)N_{T,0}}. \quad (11)$$

At thin breeding the decay-plus-loss term in the denominator consumes the surplus and the reservoir cannot double at all; only the advanced blanket ($\text{TBR}_{\text{net}} = 1.42$) opens a positive doubling window, with $t_{\text{double}} \approx 0.27 \text{ yr}$ at the frozen startup inventory (HX-08, BV-05).

Equation (11) is why tritium doubling is a blanket problem, not merely a scheduling one.

4.3 The fleet coupling and the lunar gate

Now couple the flywheel to a burner. A burner consumes helium-3 at $D_{3\text{He}} = 170.2 \text{ kg}$ per full-power year (BR-OUT-4); its neutron-lean central cell recycles $\sim 18\%$ in situ via the D–D branch (AG-CX-17). Supplying the remainder from the equilibrium breeder flywheel of equation (10), $s_{3\text{He}} = (\text{TBR}_{\text{net}} - 1)\dot{N}_{\text{burn}}$, requires a fleet ratio

$$\mathcal{N}_{\text{b/b}} = \frac{D_{3\text{He}}}{s_{3\text{He}}} = \frac{D_{3\text{He}}}{(\text{TBR}_{\text{net}} - 1)\dot{N}_{\text{burn}}} \quad (12)$$

breeders per burner. Equation (12) evaluates to $\mathcal{N}_{\text{b/b}} = 192/105/45$ at $\text{TBR}_{\text{net}} = 1.186/1.34/1.80$ (figure 5), and even at the advanced ceiling of 1.42 it is ≈ 85 . *Indigenous helium-3 therefore cannot fuel a burner fleet*: the 12.32-year tritium half-life makes the decay-sourced supply too slow by one-to-two orders of magnitude. Burner-scale helium-3 is consequently lunar-gated around 2038–40, from the solar-wind-implanted regolith resource [23, 24]. The breeder’s helium-3 remains a strategic co-product—of national-security, cryogenic and quantum-sensing value—and is explicitly *not* bulk fuel and *not* revenue; the indigenous-non-closure result and the lunar gate are carried in full by the fuel-cycle companion (doi:10.5281/zenodo.22645734).

5 The burner power balance

The second machine converts a neutron-lean D-³He plasma into electricity. It is a tandem mirror: a long central cell confined at both ends by high-field plugs that raise an ambipolar potential.

5.1 The plug potential as a Boltzmann logarithm

Electrostatic end-confinement of central-cell ions is set by the plug ambipolar potential, which for a Maxwellian electron population equilibrating along the field is a Boltzmann logarithm of the plug-to-central density ratio [14, 15],

$$\varphi_i = T_e \ln \left(\frac{n_p}{n_c} \right). \quad (13)$$

At the frozen operating point $T_e = 89.72$ keV and $n_p/n_c = 16$, equation (13) gives $\varphi_i = 89.72 \ln 16 = 248.7$ keV, reproducing the canonical anchor $\varphi_i = 248.75$ keV to three figures (figure 7). The confinement enhancement of the central cell then follows Pastukhov’s collisional-loss scaling, $\tau_{\parallel} \propto (R_m / \log R_m) (\varphi_i / T_i) e^{\varphi_i / T_i}$, with R_m the mirror ratio; the density-ratio requirement $n_p/n_c = 16$ is the burner’s one residual physics gate, and its $Q_E \geq 1$ floor sits at $n_p/n_c^* = 13.97$ (KX-L1-A21-BU). The plug-potential and sloshing-ion physics are developed in the burner companions.

5.2 Direct conversion and the engineering gain

Because the D-³He output is dominated by a 14.7 MeV proton, the burner recovers charged-particle power directly rather than through a thermal cycle. With direct fraction $f_{\text{dir}} = 0.495$, direct-converter efficiency $\eta_{\text{DEC}} = 0.49$ [22], thermal efficiency η_{th} on the residual and neutron-carried power, and recirculating electrical power P_{rec} (heating, plugs, cryogenics, balance of plant), the gross and net electric powers are

$$P_E^{\text{gross}} = \eta_{\text{DEC}} f_{\text{dir}} P_{\text{fus}} + \eta_{\text{th}} (1 - f_{\text{dir}}) P_{\text{fus}}, \quad (14)$$

$$Q_E \equiv \frac{P_E^{\text{gross}}}{P_{\text{rec}}}, \quad P_E^{\text{net}} = P_E^{\text{gross}} - P_{\text{rec}} = P_E^{\text{gross}} \left(1 - \frac{1}{Q_E} \right). \quad (15)$$

The frozen zero-dimensional balance closes at $Q_E = 1.318$ with $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW and a neutron energy fraction of only $f_n = 5.44\%$ (BU-L1-A1, KX-L1-A3-BU); a 40,000-sample global Monte-Carlo returns $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$ across the design ranges, the central-cell β_c dominating the variance (correlation 0.93; KX-L1-A5-BU, figure 6). The recirculation leverage $P_E^{\text{gross}}/P_{\text{rec}} \approx 4.14$ sets the millisecond load-following depth in the data-centre housing (MV-L1-A1). Across a device-length ladder the net electric output is +104.2/ + 850.0/ + 2831.9 MWe at 55/440/1400 m (KX-L1-A9); the recirculation-resolved Q_E closure and the direct-converter collector physics are carried by companions (doi:10.5281/zenodo.22645860). The two machines’ contrasting neutron budgets—breeder $\sim 80\%$ of fusion energy in neutrons, burner 5.44%—are compared in figure 9.

6 Computational methodology: the NVIDIA GPU HPC campaign

Every quantitative claim in this paper was produced, and then independently re-run, on the Kronos NVIDIA GPU high-performance-computing campaign. We name the codes actually exercised for the platform-level results and state the verification protocol; no economic or cost information is part of this account.

6.1 Codes and problem scale

The design set spans four solver classes. *Transport and turbulence*: the negative-triangularity quiescence underlying the breeder gain of equation (4) was established by nonlinear gyrokinetics with CGYRO [13], GPU-resident flux-tube runs at the operating gradients, cross-checked by quasilinear TGLF/TGYRO mappings. *Neutronics*: the tritium-breeding ladder of equation (6) and figure 3 was computed with OpenMC [12] on ENDF/B-VIII.0, first in a CSG radial model and then on a CAD-faithful STEP-to-DAGMC geometry with real ports and penetrations (BV-05B), each run tallying $\gtrsim 10^9$ source histories for converged three-dimensional breeding and coverage tallies. *Magnets and structure*: the current-sharing margins of the two distinct coils (§2.3, figure 8) come from REBCO $I_c(B, T, \theta)$ critical-surface scaling coupled to finite-element stress. *Reduced-order and fuel-cycle*: the surplus law (7), the flywheel ordinary differential equation (8) and the fleet ratio (12) were integrated in the Kronos toolkit with a stiff solver and swept by the

global Monte-Carlo layers, including the 40,000-sample burner uncertainty quantification of figure 6. The mirror 0-D balance of equation (15) was benchmarked against the WHAM physics basis and the Pastukhov-plug operating point [18] (doi:10.5281/zenodo.22645911).

6.2 Verification and reproducibility

The campaign was run under a verification-first protocol. Each headline quantity is a *frozen anchor*: a register serial with a byte-for-byte provenance record, re-run on an independent library or an independent node to confirm reproduction before it enters a paper (doi:10.5281/zenodo.22645710). The breeder gain, the neutron source strength, the breeding ladder, the surplus law, the flywheel constants, the plug potential and the burner gain each carry such a serial, quoted inline above. GPU acceleration enters where it is decisive—the gyrokinetic turbulence runs, the Monte-Carlo neutron transport, and the large Monte-Carlo uncertainty sweeps—while the reduced-order fuel-cycle and balance solvers run on a local node so that the analytic flywheel result is reproducible on a workstation. The numerical formulation is deliberately conservative: linear inventory balances with closed-form checks (equations 9–10), converged Monte-Carlo tallies with reported statistical error, and gyrokinetic spectra resolved to the electron scale.

7 Results

Table 1 collects the frozen platform anchors; figures 2–10 present the derived results. The breeder closes on model as a strategic-materials platform: gain $Q = 3.076$, source $3.0 \times 10^{19} \text{ n s}^{-1}$ (equation 5), and a breeding ladder that clears self-sufficiency and reaches $\text{TBR}_{\text{net}} = 1.42$. The helium-3 flywheel is fully specified by the closed-form equations (9)–(10): a $\tau = 17.8 \text{ yr}$ discharge that makes indigenous burner-fleet fuelling require 45–192 breeders per burner (figure 5). The burner closes on model at $Q_E = 1.318$ with $P(Q_E > 1) = 1.0$ over the sampled envelope, behind a buildable but distinct 26.49 T plug coil.

Table 1. Frozen platform anchors (Tier-2 canon; register serials in §§2–5). Two distinct magnets. No economics.

quantity	value	role
breeder gain Q	3.076	subcritical materials factory (eq. 4)
breeder P_{fus}	85.04 MW	source strength (eq. 5)
breeder I_p	9.66 MA	compact ST current
$R_0, A, \kappa, \delta, B_0$	1.20 m, 2.5, 2.0, -0.30 , 8 T	geometry / field
centrepost peak field	16.84 T	magnet 1 (breeder)
net TBR ceiling	1.42 (self-suff. at 1.05)	tritium ladder (eq. 6)
tritium surplus law	$(\text{TBR}_{\text{net}} - 1)\dot{N}_{\text{burn}}$	flywheel input (eq. 7)
tritium half-life	12.32 yr ($\tau = 17.8 \text{ yr}$)	flywheel constant (eq. 9)
in-situ ^3He harvest	$\sim 154 \text{ g yr}^{-1}/\text{unit}$	strategic co-product
breeders per burner	45–192	fleet non-closure (eq. 12)
burner gain Q_E	1.318	engineering gain (eq. 15)
burner f_n	5.44%	neutron-lean
plug density ratio n_p/n_c	16 ($\varphi_i = 248.7 \text{ keV}$)	residual gate (eq. 13)
plug coil peak field	26.49 T	magnet 2 (burner)
η_{DEC}	0.49	direct conversion

8 Discussion

8.1 The platform against the single-machine benchmark

Read against the standard net-electricity benchmark, the platform’s value is that it decouples *what fusion is good at now* from *what it must eventually do*. A D–T plasma at $Q = 3.076$ is far from an ignited power reactor, but it is an excellent 14 MeV source (equation 5): its breeding ladder clears self-sufficiency (figure 3) exactly where DEMO-class blanket studies place the difficulty of tritium self-sufficiency [11], and it does so in a compact spherical-tokamak volume of the kind Menard *et al.* identified as the natural fusion-nuclear-science platform [5]. The burner, at $Q_E = 1.318$ and $f_n = 5.44\%$, is quantitatively the neutron-lean converse: it sits within the advanced-fuel envelope mapped by Kulcinski and Santarius [19] and Nevins [20], respects Rider’s warning about non-equilibrium recovery by keeping the plasma near-Maxwellian and paying the recirculation honestly (equation 15) [21], and reproduces the WHAM-class Pastukhov-plug operating point [18].

8.2 Why the flywheel is the load-bearing result

The most consequential number in the paper is not a gain but a time constant. Equations (9)–(10) show that the helium-3 co-product is governed by the $\tau = 17.8 \text{ yr}$ decay of banked tritium, and

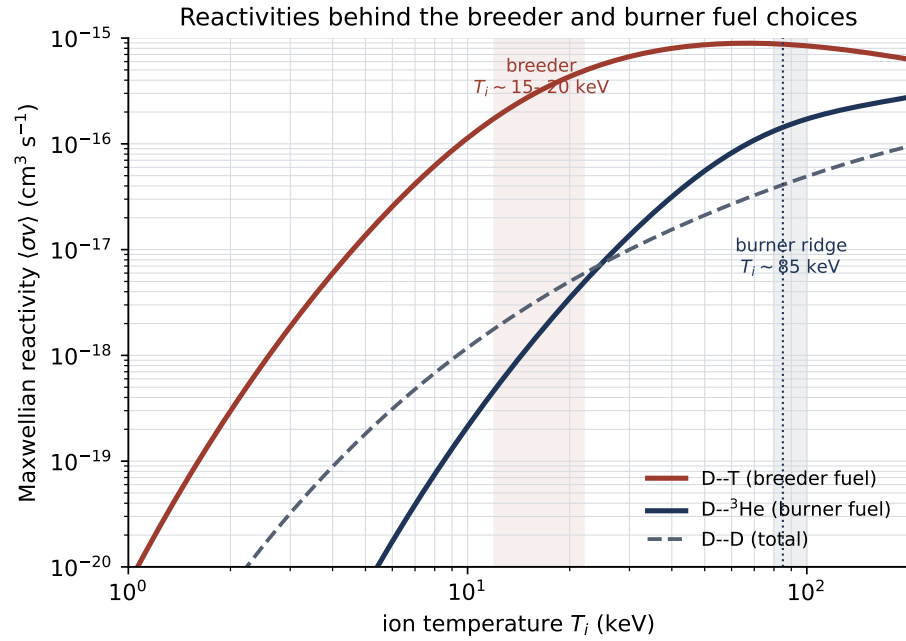


Figure 2. Bosch-Hale reactivities (equation 1) for the platform’s fuels. The breeder occupies the D–T knee at $T_i \sim 15\text{--}20\text{ keV}$; the burner occupies the D– ^3He ridge at $T_i \sim 85\text{ keV}$. Data computed from the parametrisation [3].

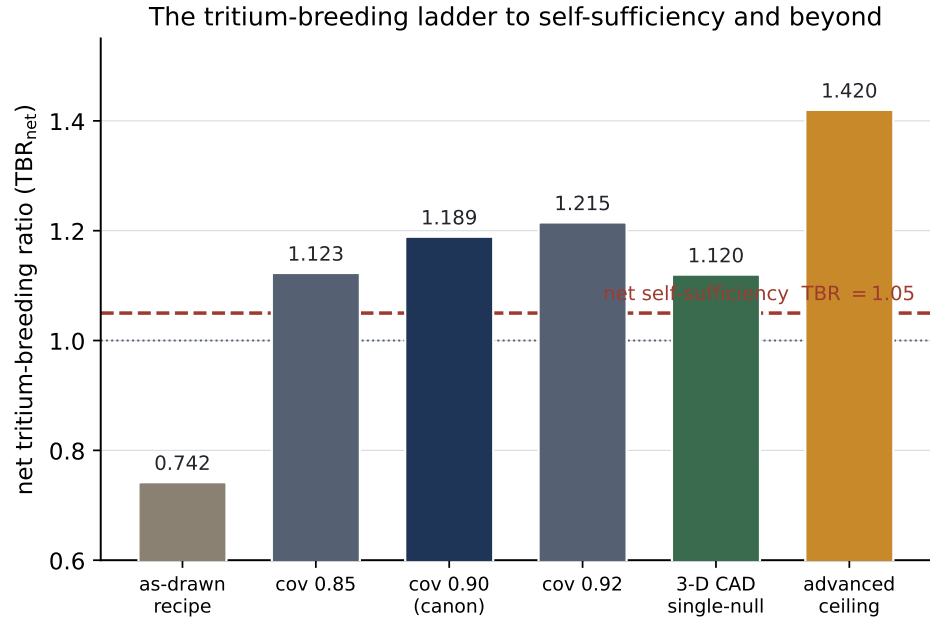


Figure 3. The net tritium-breeding ladder (equation 6): as-drawn recipe, coverage lever at $f_{\text{cov}} = 0.85/0.90/0.92$, CAD-faithful single-null three-dimensional value, and advanced-blanket ceiling. The self-sufficiency line $\text{TBR}_{\text{net}} = 1.05$ is cleared with margin. OpenMC on ENDF/B-VIII.0 (BV-05, BV-05B, BR-L2-A7u).

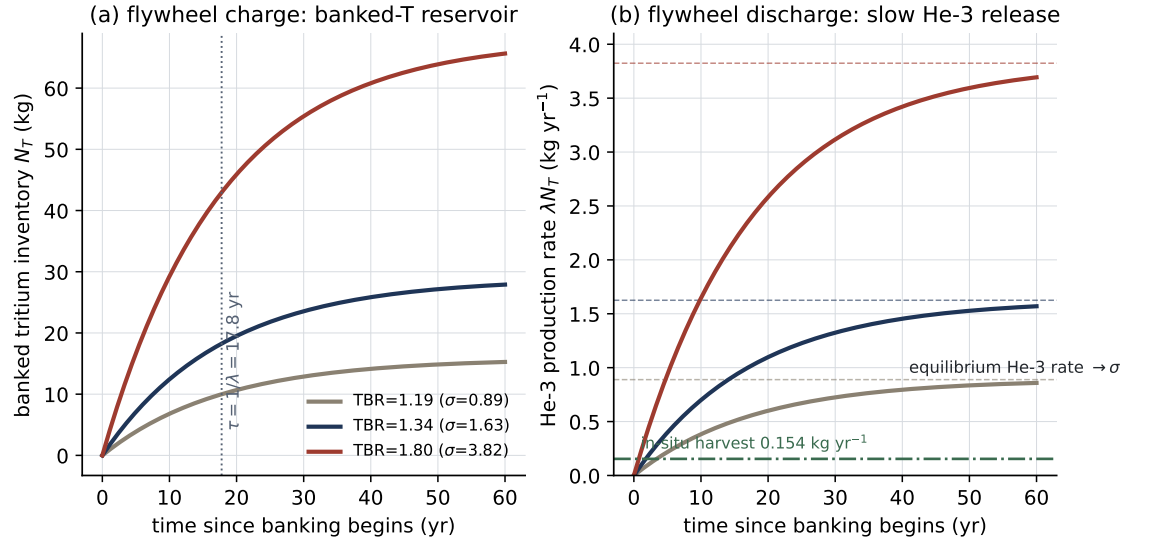


Figure 4. The helium-3 flywheel (equations 8–10). **(a)** banked-tritium inventory $N_T(t)$ charging toward σ/λ ; **(b)** helium-3 release rate $\lambda N_T(t)$ approaching the surplus σ over the e -folding time $\tau \approx 17.8$ yr, with the in-situ harvest floor. Curves shown for $\text{TBR}_{\text{net}} = 1.186/1.34/1.80$ (BR-OUT-2, BR-OUT-4).

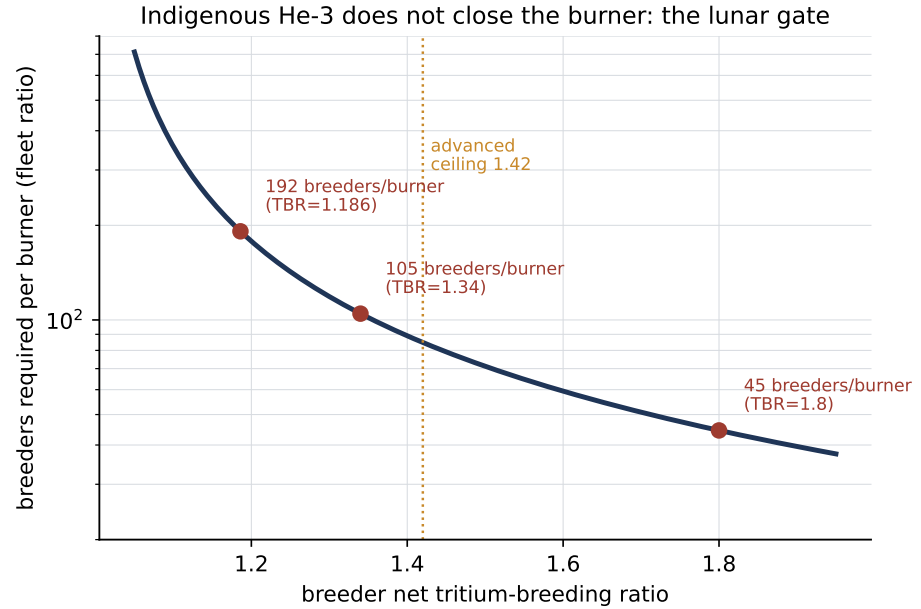


Figure 5. The fleet coupling (equation 12): breeders required per burner versus breeder net TBR, for a burner demand $D_{3\text{He}} = 170.2 \text{ kg fpy}^{-1}$. Points mark the frozen anchors 192/105/45; even the advanced ceiling (1.42) needs ≈ 85 . Indigenous helium-3 does not close the burner—the lunar gate (BR-OUT-4).

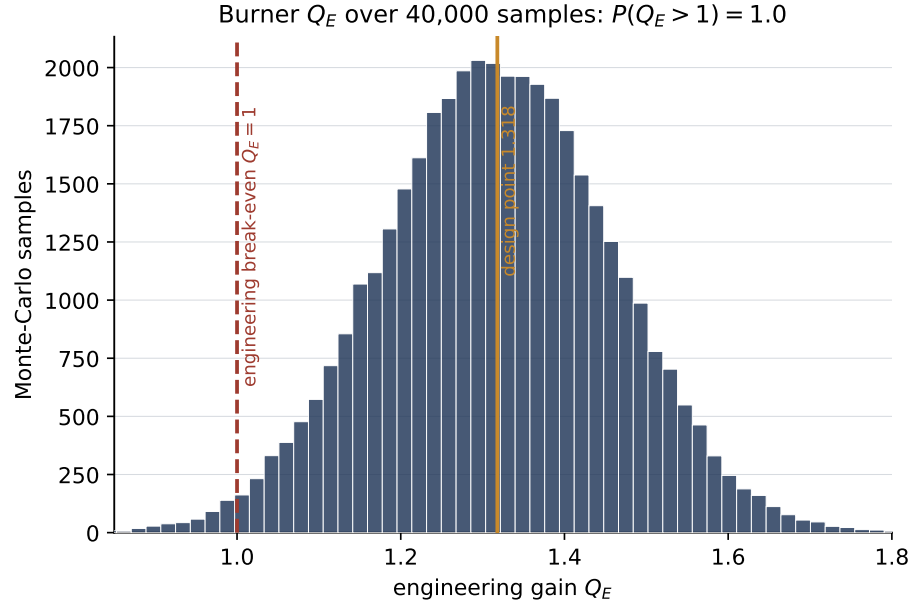


Figure 6. Burner engineering gain over the 40,000-sample global Monte-Carlo (KX-L1-A5-BU): $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$; design point 1.318. Variance dominated by central-cell β_c (correlation 0.93).

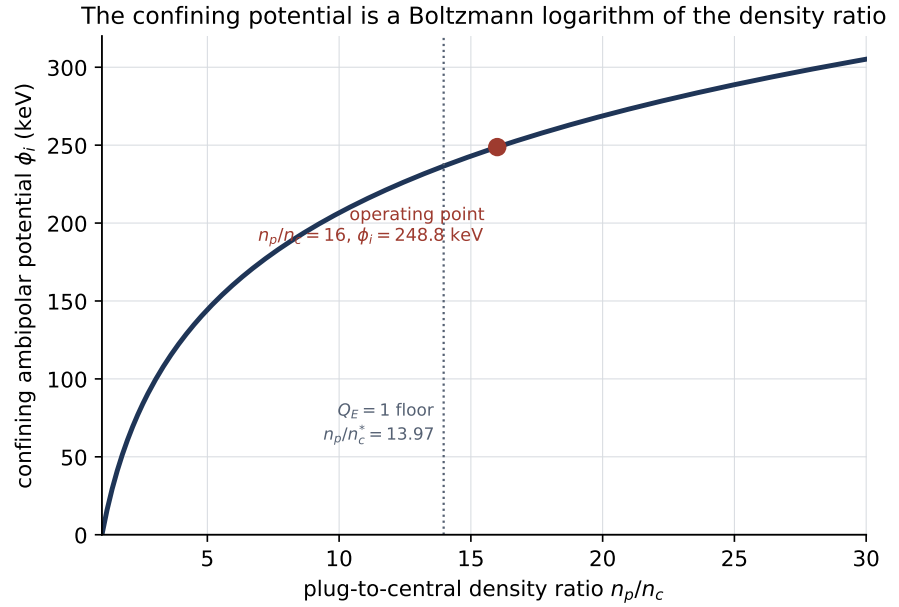


Figure 7. The confining ambipolar potential as a Boltzmann logarithm of the density ratio (equation 13) at $T_e = 89.72$ keV. The operating point $n_p/n_c = 16$ gives $\phi_i = 248.7$ keV, reproducing the canonical anchor; the $Q_E=1$ floor sits at $n_p/n_c^* = 13.97$.

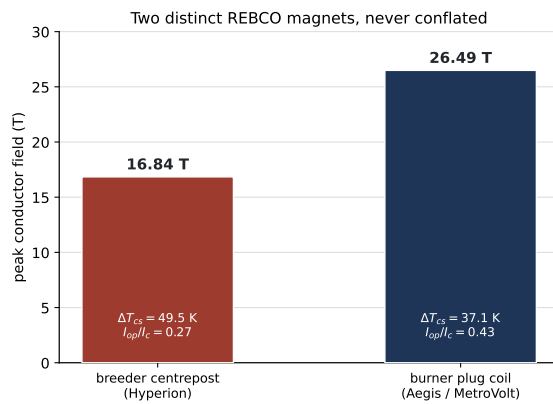


Figure 8. The two distinct REBCO magnets: breeder centrepost at 16.84 T and burner plug coil at 26.49 T, with current-sharing margins and load fractions annotated. Never conflated.

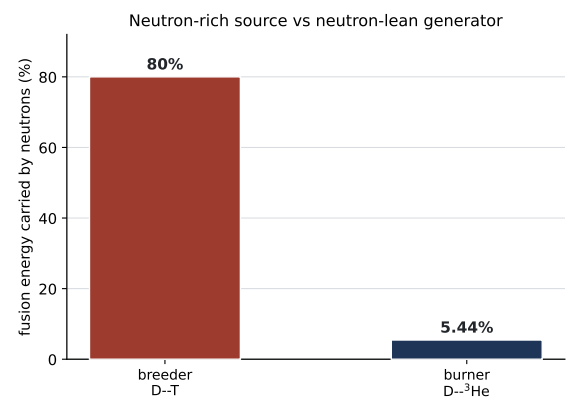


Figure 9. Neutron energy fraction: the breeder is a deliberately neutron-rich D-T source ($\sim 80\%$), the burner a neutron-lean D-³He generator (5.44%). The contrast underlies the siting split.

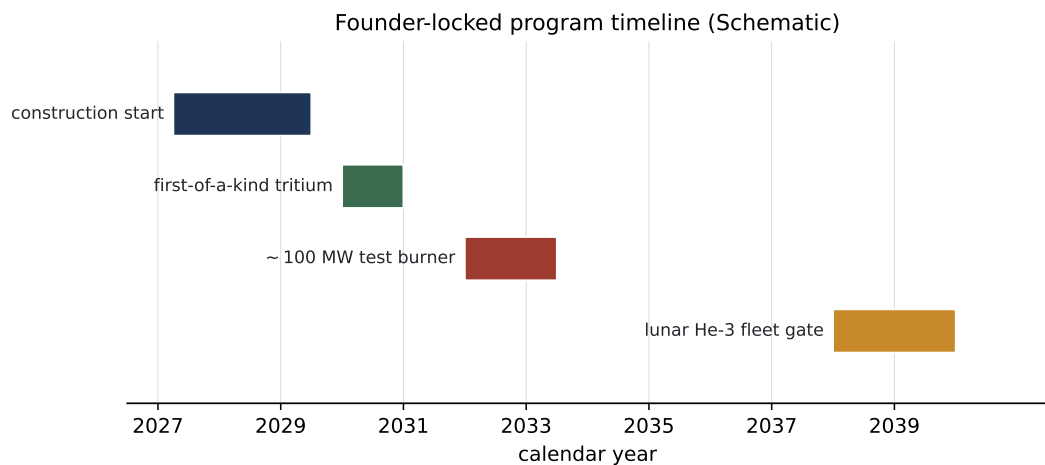


Figure 10. (Schematic.) The founder-locked program timeline: construction start (Q2 2027), first-of-a-kind tritium (2030–31), the ~ 100 MW test burner (2032), and the lunar-helium-3 fleet gate (2038–40). Positions are program milestones, not measured data.

equation (12) turns that slowness into the flat statement of figure 5: 45–192 breeders would be needed to fuel one burner from indigenous decay. This is why the platform’s helium-3 is a strategic *co-product* and burner-scale fuel is *lunar-gated* [23, 24]—a conclusion that falls straight out of the half-life and is robust to every plausible breeding assumption, since even the 1.42 ceiling needs ≈ 85 . The honest framing throughout is that helium-3 value is not helium-3 revenue and the loop does not close between the two machines; the fuel-cycle companion carries the full indigenous-non-closure case (doi:10.5281/zenodo.22645734).

8.3 Two magnets, one basis

A recurring source of confusion in compact high-field concepts is the magnet. The platform shares a REBCO *basis*—no-insulation coated conductor, demountable practice, quench-margin methodology—but uses two *distinct* magnets: the 16.84 T breeder centrepost and the 26.49 T burner plug coil (figure 8). Both sit within demonstrated REBCO field practice [8], but they are different devices with different load lines, and the platform’s magnet-integrity case depends on never conflating them.

9 Limitations

One honest gate bounds the burner side of the platform. The burner’s engineering gain $Q_E = 1.318$ closes on model with $P(Q_E > 1) = 1.0$ over the sampled envelope, but its one residual *physics* gate is the plug potential: the requirement $n_p/n_c = 16$ (equation 13) is a density-ratio target that a plug-scale potential experiment must demonstrate at the operating T_e , and the 40,000-sample uncertainty quantification fixes the plug density and excludes central-cell radial transport (KX-L1-A5-BU). The magnet is *not* the gate—the 26.49 T plug coil qualifies structurally (doi:10.5281/zenodo.22645915)—and the flywheel result is a closed-form consequence of a measured half-life. The distance from closed-on-model to closed-demonstrated for the burner is therefore one named experiment, the plug-potential demonstration, not open physics.

10 Conclusion

Kronos is one platform with two machines and a single frozen physics record. The breeder is a compact negative-triangularity spherical tokamak at $Q = 3.076$ and $P_{\text{fus}} = 85.04$ MW that makes strategic materials—tritium along a net breeding ladder to a 1.42 ceiling (clearing self-sufficiency at 1.05), helium-3, 14 MeV neutrons, and co-produced isotopes—behind a 16.84 T centrepost. The burner is a D–³He tandem mirror at $Q_E = 1.318$ and a neutron energy fraction of 5.44%, recovering charged-particle power at $\eta_{\text{DEC}} = 0.49$ behind a distinct 26.49 T plug coil. Linking them is the helium-3 flywheel, whose governing balance (equations 8–10) shows that banked tritium discharges to helium-3 over an e -folding time of 17.8 yr—slow enough that indigenous helium-3 cannot fuel a burner fleet (equation 12; figure 5), which is why burner-scale fuel is lunar-gated and the breeder’s helium-3 is a strategic co-product rather than bulk fuel. Stated in two tiers: all three products close on the frozen physics, every component is buildable within demonstrated REBCO and mirror practice, and the distance to demonstrated closure is a small number of named experiments—the measured breeding ratio at first-of-a-kind operation and the plug-potential demonstration—not open physics.

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Reproducibility. Every headline quantity in this paper is a frozen anchor in the Kronos Physics De-Risking Register [25] (doi:10.5281/zenodo.22133057), quoted inline by serial. The flywheel model (equations 8–10), the surplus law (7) and the fleet ratio (12) are regenerable from the archived toolkit and the figure-generation script deposited with this paper; the breeding ladder and burner uncertainty quantification are regenerable from the archived OpenMC and toolkit inputs. This paper is archived at doi:10.5281/zenodo.22645764; official version: <https://www.kronosfusionenergy.com/publications/integrated-platform-overview.html>.

Scope — what this paper does not claim. No product is claimed *demonstrated*-closed; each closes on model, with the distance to demonstrated closure stated as named experiments. No economic, cost, or valuation content appears anywhere in this paper, by design.

Data availability

The reactivity, breeding-ladder, flywheel, fleet-ratio, plug-potential and burner-uncertainty data plotted in figures 2–6 and 7–9 are archived with the deposit at doi:10.5281/zenodo.22645764, which references the Kronos 2026 series including the tritium-breeding (doi:10.5281/zenodo.22645746), net-TBR fleet (doi:10.5281/zenodo.22645706), helium-3 fuel-cycle (doi:10.5281/zenodo.22645734), burner power-balance (doi:10.5281/zenodo.22645860) and plug-coil (doi:10.5281/zenodo.22645915) companions. All quantitative values trace to the register [25] by the serial identifiers cited inline.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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